Additive Manufacturing: Prototype to Production 162

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Class Outline

Additive Manufacturing and Production
AM Production Processes
AM Production Advantages
AM Production Disadvantages
AM Production Review
Common AM Applications
AM Material Standards
AM Material Considerations
Final Review

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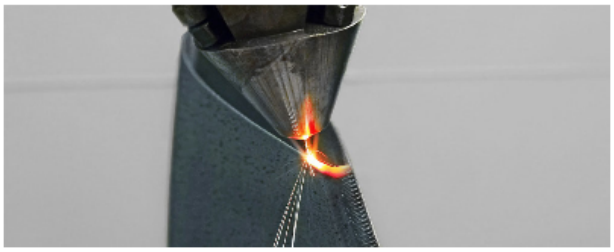
Lesson: 1/9

Additive Manufacturing and Production



Additive manufacturing (AM) includes multiple processes that build parts by fusing layers of raw material. Although it is typically associated with rapid prototyping, a growing number of manufacturers are working with AM to create end-use products and near net-shape, or near-finished parts. End-use is a term used to describe parts and products used directly by the consumer. These includes things like custom medical equipment, such as prosthetics, and printed circuit boards (PCBs) for electronics. Manufacturers can also use AM for repairs, such as filling cracks or holes in existing parts.

Increasingly, AM processes are being combined with traditional methods to create hybrid manufacturing, which builds a completed part with greater efficiency. In some cases, a manufacturer may use AM to make customized fixtures and jigs for traditional processes. Additionally, utilizing AM techniques to make small-batch lots can save manufacturers the cost of creating tooling, workholding, and automation for limited-run parts. Using AM processes for production may have some disadvantages, but compared to traditional methods, AM has proved to be more agile for meeting customer demands.

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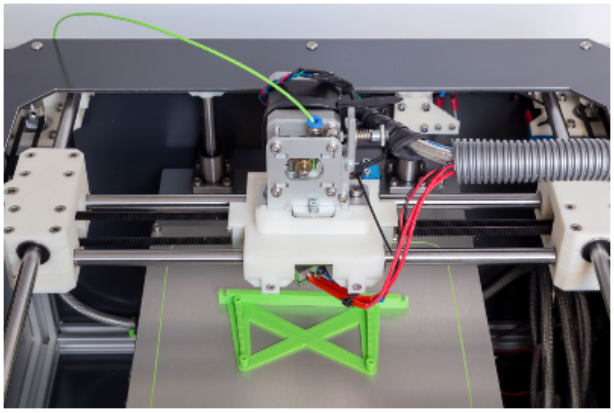
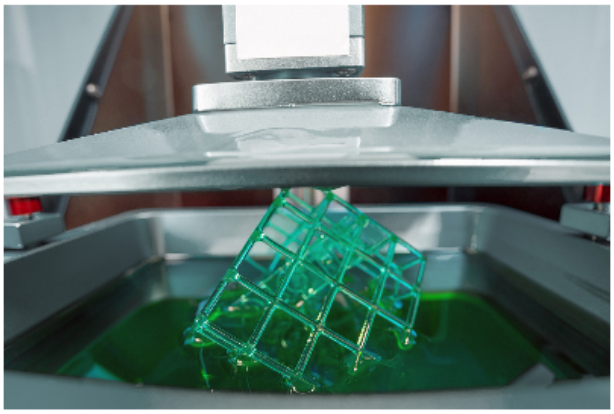
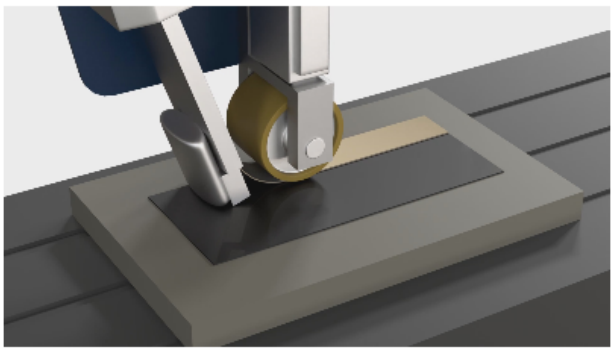
AM Production Processes



Directed energy deposition (DED), material extrusion, and material jetting use a device such as a nozzle to deposit build materials.



Vat photopolymerization directs an ultraviolet (UV) light to cure selected layers of specialized resins to form parts.



Sheet lamination bonds sheets of material together using heat or other means, with a feed roller.

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Powder bed fusion (PBF) and binder jetting processes bond selected areas of material in a powder bed. Each of these methods also includes variations, which are often based on the build materials. For example, binder jets may use metal powder to make stainless steel parts for airplanes or ceramic powder to create parts like a high-pressure turbine nozzle.

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AM Production Advantages

Additive manufacturing eliminates many of the limitations of traditional machining processes that subtract or remove material. Subtractive machining requires a different machine setup for every part or prototype, which can take a significant amount of time to change. Machining is also limited to the shape of the raw material and the abilities of a tool. For example, a facility may need to create a part that requires two components to meet at a 90° angle. Using a machining process would require either removing and wasting a large amount of material between the components, or creating them separately and using a joining process like welding, which would add a large amount of time to the part production process. AM production can reduce waste by building the angled part in one piece, using only the necessary amounts of raw material.

In rapid prototyping, AM can also enable manufacturers to quickly create iterative design prototypes that focus on functionality. These prototypes can have varying material compositions layer by layer, or material complexity and allow for hierarchical complexity, which lets a multi-scale part transition from simple to complex shapes as it is built. The ability to quickly create many test versions of a part enables manufacturers to meet customer demands with greater speed and agility compared to traditional processes. For example, an aircraft manufacturer may need to quickly develop a new type of duct component to prevent a front windshield from fogging up and impairing pilot visibility. Using a PBF process like direct metal laser sintering (DMLS), they can rapidly create and test multiple prototypes to customize the end-use part to meet the unique design needs or to meet strict aircraft requirements. They can then use the same DMLS process to create the end-use part.

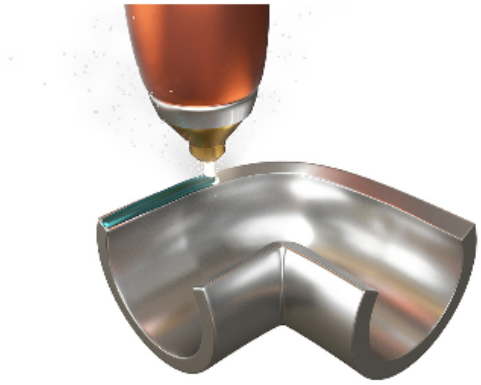

Figure 1: Additive Process: Directed Energy Deposition

Figure 2: Traditional Processes: Drilling and Welding

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AM Production Disadvantages

Adopting additive manufacturing requires significant up-front investments in machines and materials, though the total cost of investment can be lower than traditional manufacturing due to reduced material, setup, and other costs. Working with AM can also require specialized, multi-functional training. For example, an operator building thermoplastic components using material extrusion must understand the material they are working with, the material extrusion process, and machine maintenance requirements such as cleaning extrusion nozzles. Similarly, operators must understand that AM processes that produce low-strength parts may not be appropriate for building structural components like a bracket for a door latch.

End-use products made with additive manufacturing may also need post-processing, or extra steps, before they are completed. These extra steps can lower build rates, or the number of parts made during a set period. For example, some AM parts require support structures to support the part during building, and personnel often need to remove these structures manually. In other instances, meeting the precise design size and specifications, or dimensional accuracy, for an AM part may prove challenging. To compensate, a manufacturer may need to use subtractive post-processing to drill a hole or grind a smooth surface finish.

An increasingly common method of avoiding the disadvantages of both additive and subtractive processes is hybrid manufacturing. Hybrid manufacturing combines AM with traditional computer numerical control (CNC) machines, either by creating a specialized hybrid machine or by adding an AM nozzle to an existing CNC machine. Hybrid processes allow manufacturers to create some part features using AM and others using traditional machining without needing multiple machines or setups.

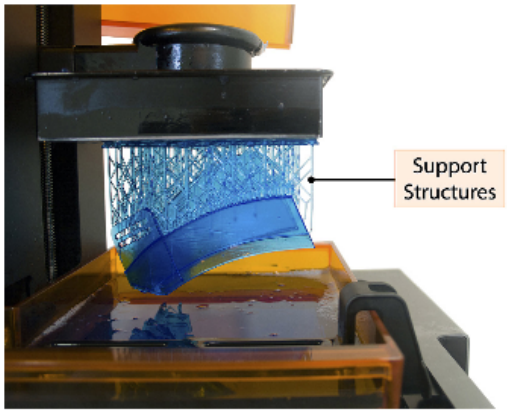

Figure 1:

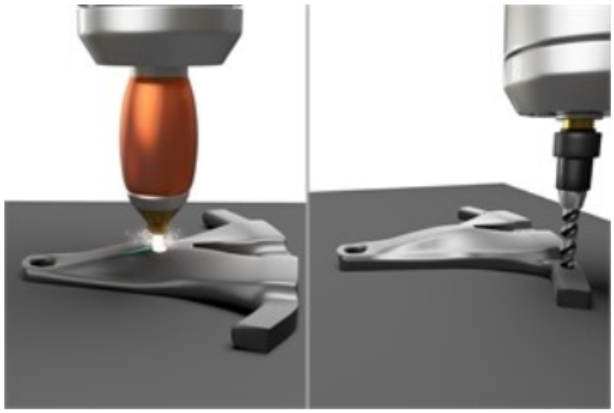

Figure 2: Hybrid Manufacturing

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Lesson: 5/9

AM Production Review

- **Support Structures**: Must often be removed manually from parts built with additive manufacturing.
- **Iterative Design**: Typically requires many different variations to refine a part prototype.
- **AM Processes**: Builds parts by joining materials together, layer by layer, based on a 3D computer design.
- **End-Use Products**: Parts and products used directly by the consumer.

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Common AM Applications

A complex product may consist of a few AM components among traditionally made products. For example, an automaker may use fabrication processes to create the majority of a car door but use AM to make a window guard rail that is difficult to produce with traditional methods.

AM is ideal for creating custom medical equipment and devices. Additive processes are commonly used to make devices including hearing aids, prosthetics, and dental products. Manufacturers also use AM processes to make different kinds of orthopedic implants for the spine and other areas of the body.

AM used for small-batch part lots can be cost effective, since additive manufacturing processes do not require different setups for each part lot and may prevent needing to use multiple processes. For example, using AM may save manufacturers from making non-standard tooling and workholding devices to create a small number of customized parts.

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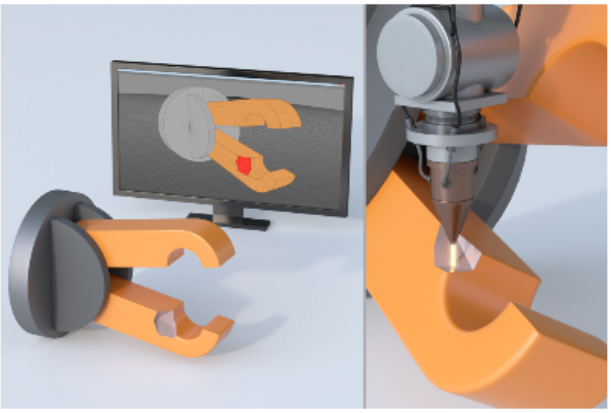

Additive technology repairing cracks.

Uses for additive manufacturing are growing every day as the technology evolves. For example, laser additive technology is used to repair cracks in metal components. In addition, manufacturers can also use AM to make custom tools that assist with traditional manufacturing. These can include things like a fixture to hold parts securely or a mold used to shape a specialized part.

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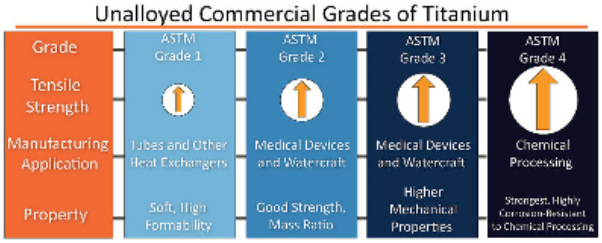
AM Material Standards



AM methods use a variety of materials. The most typically recognized AM process is 3D printing, which is generally material extrusion using plastic resin as the raw material. However, AM processes that use metal, like DED and PBF, are common for industrial production. Other popular materials used in additive manufacturing include polylactic acid (PLA), ceramics, and nylon.

During the evolution of AM technology, many AM developers used their own terms and standards, which caused confusion. The International Organization for Standardization (ISO) and the American Society for Testing and Materials (ASTM) work together to create a framework of unified standards to ensure all AM business products have the same high quality worldwide. The AM specifications for creating certain metal grades, or categories, of titanium and aluminum is one of the more important standards created by the ASTM/ISO collaboration. For example, these grades can provide guidelines for AM manufacturers who build castings for surgical implants or dental restorations. The unified set of standards enables manufacturers to communicate better with each other. In turn, this ensures that anyone using an additively manufactured product, whether as a passenger on an aircraft or within their own body, can trust the product's quality.

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AM Material Considerations

AM end-use products can require different inspection methods than traditional manufactured parts because they are built differently. For instance, many AM parts may have more complicated internal structures compared to traditional parts, which are often machined from solid pieces of raw material.

Manufacturers inspect parts to prevent serious defects, or flaws that prevent a part from meeting specifications for a part. Destructive testing and nondestructive testing represent two methods of inspecting parts for defects. Destructive testing includes taking a random sample from a production line and inspecting it in a way that breaks or damages the part, which turns it into scrap. For example, a manufacturer using AM techniques to mass produce a part may use tensile testing, which stretches the part until it breaks, to determine its strength. Nondestructive testing includes inspection methods that do not damage a part or its future usability. For example, a manufacturer may decide to use a nondestructive testing method like ultrasonic inspection, which uses sound, or x-ray testing, which uses light.

Not only can storage and handling affect the quality of the built part, but it can also impact production times. In addition, AM techniques using metal must maintain the quality of the powder to consistently grade quality standards set forth by ASTM and ISO.

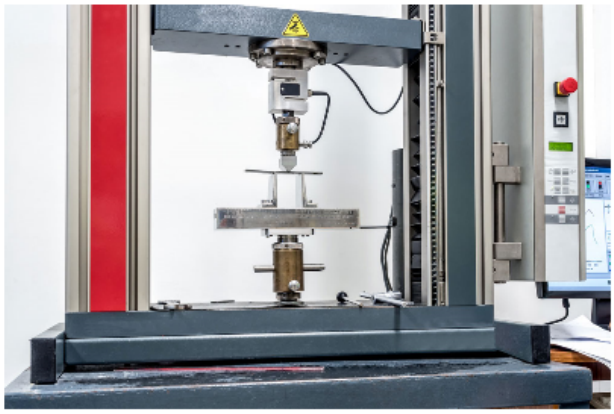

Figure 1: Destructive Tensile Testing

Figure 2: Nondestructive X-Ray Testing

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Lesson: 9/9

Final Review
Which of the following describes a common end-use production strategy for AM?

- **Small product batches**
- Large product batches
- Traditional manufacturing
- Subtractive manufacturing

Which of the following is an organization that creates material standards for AM developers?

- **ASTM**
- PLA
- PBF
- DED

Which of the following describes a form of destructive testing?

- **Bending a part to verify its strength.**
- X-ray examination of a part for defects.
- Ultrasonic inspection of a part for cracks.
- Measuring a manufactured part with calipers.

What is the term used to describe the category, rank, or level of quality?

- **Grade**
- Casting
- Resin
- Defect

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Class Vocabulary

3D Printing	three-dimensional printing. The process of producing a 3D object using a specialized tool that creates successive layers of material. 3D printing must technically involve the use of a print head or nozzle, but the term is often used interchangeably with additive manufacturing (AM).
Additive Manufacturing	AM. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.
Aluminum	A lightweight metal that is silvery white in color. Aluminum resists corrosion and is a good conductor of electrical and thermal energy.
AM	Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. AM methods typically build up layers of material to create an object.
American Society For Testing And Materials	ASTM. An organization that writes and updates standards for a broad range of materials, including metals. The American Society for Testing and Materials has created seven classifications for additive manufacturing (AM) processes.
ASTM	The American Society for Testing and Materials. An organization that writes and updates standards for a broad range of materials, including metals. ASTM has created seven classifications for additive manufacturing (AM) processes.
Binder Jetting	An additive manufacturing (AM) method in which liquid binder is used to join powdered materials to create parts. Binder jetting can create parts out of metal, polymer, or ceramic.
Bracket	A mounted support device. Brackets can be optimized through additive manufacturing (AM) methods as they impose fewer design limitations than traditional manufacturing operations.
Build Rates	The time it takes to build a part or the number of parts that can be made in a set period of time. Build rates for additive manufacturing (AM) are considerably lower than traditional manufacturing.
Computer Numerical Control	CNC. A programmable system of software and hardware that directs the operation of a machine. Computer numerical control systems use mathematical data to direct machine movements.
Customize	Created specially for a specific machine or customer. Customization is a key advantage of additive manufacturing (AM).
DED	Directed energy deposition. An additive manufacturing (AM) process in which focused thermal energy is used to melt materials as they are fed or blown through a nozzle. DED is often used with powdered or wire metal.
Defects	An imperfection in a part that prevents it from operating correctly. Defects sometimes appear in additive manufacturing (AM) parts when the layers do not adhere to each other correctly.
Destructive Testing	A category of inspection processes that evaluate a part's properties and performance using methods that lead to part failure. Destructive testing renders a part unusable.
Dimensional Accuracy	The ability of a part to meet the size and specifications of the intended design. Dimensional accuracy is essential for maintaining a repeatable target size when manufacturing a part.
Direct Metal Laser Sintering	DMLS. An additive manufacturing (AM) process that uses a laser to bond successive layers of material in a bed of powdered metal. Direct metal laser sintering can produce complex metal parts, though they often require extensive post-processing.
Directed Energy Deposition	DED. An additive manufacturing (AM) process in which focused thermal energy is used

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to melt materials as they are fed or blown through a nozzle. Directed energy deposition is often used with powdered or wire metal.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.

Fabrication A general category of processes that produce parts by adding or joining material. Fabrication includes welding and assembly processes.

Fixtures A customized workholding device used on machine tools to position and hold a part during various machining operations. A fixture is built to hold a specific part design.

Grades A category, rank, or level of quality. Grades of materials are typically categorized by the specific amount of elements included in the materials.

Hierarchical Complexity The amount of variation allowed in the ordering of shape or material composition of a part. Hierarchical complexity allows for numerous and rapid changes in the shape and composition of a multi-scale part.

Hybrid Manufacturing A process that uses both traditional and additive manufacturing (AM) to create a finished part. Hybrid manufacturing can involve either using a traditional manufacturing process on a mostly additively manufactured part or vice versa

Inspection The examination of a product during or after its creation to confirm that it adheres to specifications. Inspection allows manufacturers to identify and correct product defects.

International Organization For Standardization ISO. An organization based in Switzerland that develops and publishes standards for various aspects of manufacturing and industry. The International Organization for Standardization is responsible for standards across a number of different industries.

ISO International Organization for Standardization. An organization based in Switzerland that develops and publishes standards for various aspects of manufacturing and industry. ISO is responsible for standards across a number of different industries.

Iterative Design Process of repeatedly testing, inspecting, and improving a prototype. Iterative design typically requires many different variations to refine a part design before putting it into full production.

Jigs A customizable, modular workholding device that supports, locates, and clamps a workpiece and also directs the cutting tool. Jigs are often used to hold irregular parts and generally decrease machine tool setup time as they provide guidance for tools.

Joining Bringing two separate materials together through some type of forming. Joining is one of the four main ways plastics can be formed.

Machining A subtractive manufacturing process that involves removing material to form an object. Machining includes methods that remove metal using cutting tools, such as milling, turning, and drilling.

Material Complexity In additive manufacturing, refers to build substances that can be processed specifically layer by layer. Material complex parts can differ in material composition.

Material Extrusion An additive manufacturing (AM) method that uses a nozzle to dispense material, usually a thermoplastic filament, onto a build platform. Material extrusion is sometimes referred to as fused deposition modeling (FDM).

Material Jetting An additive manufacturing (AM) method in which droplets of build material are selectively deposited onto a build platform. Material jetting systems use a photopolymer that is cured by ultraviolet (UV) light.

Milling A cutting operation in which a rotating multi-point cutting tool is fed along the surface of a part to remove material. Milling operations generally produce flat surfaces.

Near Net-Shape Having dimensions that are close to a final form after initial manufacturing steps. Near-net shape parts require little subsequent machining.

Nondestructive Testing A category of inspection processes that evaluate a part's properties and performance using methods that do not damage or permanently alter the part. Nondestructive testing methods can be used to locate discontinuities and defects in parts made by additive manufacturing (AM) processes.

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Nylon	A semicrystalline thermoplastic that gains strength when the fibers are stretched. Nylon is a commonly used term for polyamides (PA).
Orthopedic	Relating to the branch of medicine that focuses on the musculoskeletal system. Orthopedics often uses additive manufacturing to develop implants and prosthetics.
PBF	Powder bed fusion. An additive manufacturing (AM) method that uses thermal energy to fuse together layers of powdered polymer, metal, ceramic, or other material. PBF includes several processes, such as selective laser sintering (SLS) and direct metal laser sintering (DMLS).
Polylactic Acid	PLA. A biodegradable thermoplastic in the polyester family. Polylactic acid is a natural polymer derived from corn starch and other renewable resources.
Post-Processing	A procedure used to clean, improve, or otherwise finish a part after it has been manufactured. Post-processing procedures include abrasive finishing, heat treatment, and painting.
Powder Bed Fusion	PBF. An additive manufacturing (AM) method that uses thermal energy to fuse together layers of powdered polymer, metal, ceramic, or other material. Powder bed fusion includes several processes, such as selective laser sintering (SLS) and direct metal laser sintering (DMLS).
Printed Circuit Boards	PCBs. A layered construction of material used to mechanically support and electrically connect electronic components. Printed circuit boards use conductive pathways, or traces, etched from copper sheets laminated onto a non-conductive surface.
Prosthetics	An artificial implant, device, or limb that replaces a missing body part. Prosthetics may also refer to the design, fabrication, and fitting process involved with these devices.
Rapid Prototyping	A product development technique in which additive manufacturing (AM) methods are used to create prototypes for a traditional manufacturing operation. Rapid prototyping allows engineers to quickly create a number of prototypes in a short time period, reducing lead time.
Raw Material	An unprocessed or lightly processed object or substance used to make a finished part. Raw materials in manufacturing include metal, plastic, and ceramic.
Resin	A raw polymer, usually in the form of beads or pellets, that is not yet molded into its final shape. Resin is melted to form plastic parts.
Resins	A raw polymer, usually in the form of beads or pellets, that is not yet molded into its final shape. Resins are melted to form plastic parts.
Setup	All the necessary preparation of tooling and fixturing that occurs on a machine before operation begins. Proper machine setup is an important factor in optimizing tool life.
Sheet Lamination	An additive manufacturing (AM) process that forms an object by bonding sheets of material together using an adhesive, heat, or pressure. Sheet lamination processes melt thin sheets of material together, bonding them layer by layer, to form a single three-dimensional object.
Specifications	A description of the essential physical and technical properties of a finished part. Specifications outline important information including finished part dimensions and how the part must respond to forces acting upon it.
Support Structures	Additional material used in the additive manufacturing (AM) process to support the product as it is being constructed. Support structures are removed from the AM product as part of the additive manufacturing process.
Tensile Testing	A test that evaluates a material's tensile strength by stretching a specimen until it breaks. Tensile testing is sometimes called tension testing.
Thermoplastic	A group of plastics that can be repeatedly heated, cooled, and shaped. Thermoplastics are often used in material extrusion additive manufacturing (AM) processes.
Three-Dimensional	3D. Occurring within three planes of space. Three-dimensional objects have length, height, and width.
Titanium	A nonferrous metal that is lightweight corrosion resistant and has a high strength-to-weight ratio. Titanium is often used in the aerospace industry.
Total Cost Of Investment	All the direct and indirect expenses associated with a purchase. Total cost of

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Total Cost of Investment	All the direct and indirect expenses associated with a purchase. Total cost of investment takes into account several factors including reliability, maintenance time, and replaceability in addition to initial cost.
Turning	A cutting operation that rotates a cylindrical workpiece while gradually passing the cutting tool along the surface of the rotating part. Turning is performed on a lathe and is commonly used to create cylindrical parts.
Ultrasonic Inspection	A nondestructive testing method that sends high-frequency sound waves through a material to locate discontinuities. Ultrasonic inspection records data that the inspector can then interpret.
Ultraviolet	UV. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. Ultraviolet light is used to selectively harden a photopolymer in vat photopolymerization processes.
Vat Photopolymerization	An additive manufacturing (AM) process that traces a focused light beam over a layer of photopolymer to selectively cure it. Vat photopolymerization is often used for making softer parts that require a high degree of customization, such as dental aligners or hearing aids.
X-Ray Testing	A nondestructive testing method in which the use of an x-ray machine to examine the interior of a completed part. X-ray testing is expensive and time-consuming and therefore is typically only used in situations where part must be perfect.

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